

Maximiliano Isi

msi2114@columbia.edu | maxisi.github.io

EMPLOYMENT	Assistant Professor , Department of Astronomy, Columbia University	2025–present
	Associate Research Scientist , CCA, Flatiron Institute	2025–present
	Flatiron Research Fellow , CCA, Flatiron Institute	2021–2025
	NASA Einstein Fellow , Massachusetts Institute of Technology	2019–2021
EDUCATION	Ph.D. in Physics , California Institute of Technology Thesis Advisor: Alan J. Weinstein	June 2019 Pasadena, CA
	B.S. in Physics and Mathematics , Loyola Marymount University Thesis Advisor: Jonas R. Mureika Trustee Scholarship <i>Summa cum laude</i>	May 2014 Los Angeles, CA
KEYWORDS	Gravitational waves, black holes, general relativity, astrophysics, cosmology, statistics, data analysis, inference, hierarchical Bayesian models, machine learning.	
RANKINGS	h-index	publications
	113 / 37	259 / 68
		121k / 5k
	Citable records per InspireHEP; first number includes collaboration papers, second excludes them (see enclosed publication list). Last updated: 2026-09-25.	
FEATURED PUBLICATIONS	Top-cited short-authorlist papers on InspireHEP (collaboration and review articles excluded; mentees underlined)	
	1. M. Isi, M. Giesler, W. M. Farr, M. A. Scheel, and S. A. Teukolsky, <i>Testing the no-hair theorem with GW150914</i>, Phys. Rev. Lett., 123 (2019), p. 111102. [arXiv:1905.00869] 559 citations. 2. M. Giesler, M. Isi, M. A. Scheel, and S. Teukolsky, <i>Black Hole Ringdown: The Importance of Overtones</i>, Phys. Rev. X, 9 (2019), p. 041060. [arXiv:1903.08284] 404 citations. 3. M. Isi and W. M. Farr, <i>Analyzing black-hole ringdowns</i>, (2021). [arXiv:2107.05609] 201 citations. 4. M. Isi, W. M. Farr, M. Giesler, M. A. Scheel, and S. A. Teukolsky, <i>Testing the Black-Hole Area Law with GW150914</i>, Phys. Rev. Lett., 127 (2021), p. 011103. [arXiv:2012.04486] 165 citations. 5. G. Vajente, Y. Huang, M. Isi, J. C. Driggers, J. S. Kissel, M. J. Szczepanczyk, and S. Vitale, <i>Machine-learning nonstationary noise out of gravitational-wave detectors</i>, Phys. Rev. D, 101 (2020), p. 042003. [arXiv:1911.09083] 154 citations.	
	Top-cited LIGO Collaboration papers featuring my work (further contributions detailed below)	
	1. B. P. Abbott and others, <i>GW170814: A Three-Detector Observation of Gravitational Waves from a Binary Black Hole Coalescence</i>, Phys. Rev. Lett., 119 (2017), p. 141101. [arXiv:1709.09660] 2412 citations. I contributed the polarization analysis and formed part of the paper writing team. 2. R. Abbott and others, <i>GW190521: A Binary Black Hole Merger with a Total Mass of $150M_{\odot}$</i>, Phys. Rev. Lett., 125 (2020), p. 101102. [arXiv:2009.01075] 1620 citations. I contributed the flagship parameter estimation analysis and contributed to the writing. 3. B. P. Abbott and others, <i>Tests of General Relativity with the Binary Black Hole Signals from the LIGO-Virgo Catalog GWTC-1</i>, Phys. Rev. D, 100 (2019), p. 104036. [arXiv:1903.04467] 1132 citations. I formed part of the paper writing team and contributed the polarization analysis. 4. R. Abbott and others, <i>Tests of general relativity with binary black holes from the second LIGO-Virgo gravitational-wave transient catalog</i>, Phys. Rev. D, 103 (2021), p. 122002. [arXiv:2010.14529] [Erratum: Phys. Rev. D 112, 049902 (2025)] 1130 citations. I chaired the paper and contributed multiple analyses, including the hierarchical population, residuals and dispersion analyses. 5. R. Abbott and others, <i>Tests of General Relativity with GWTC-3</i>, Phys. Rev. D, 112 (2025), p. 084080. [arXiv:2112.06861] 1088 citations. I contributed the infrastructure for combining observations from multiple events.	

LEADERSHIP &
SERVICE**Scientific Collaborations**

LIGO Scientific Collaboration (2012–present); LISA Consortium (2019–present).

I have led or co-led multiple flagship LIGO-Virgo-KAGRA publications, including the detection of **GW250114**, the astrophysical implications of **GWTC-5** and tests of general relativity with **GWTC-2**. *As a Paper Chair or Project Manager, I have overseen the end-to-end production of results and manuscripts, coordinating the work of dozens of LVK members.* (See publication list [below](#).)

Journal and Grant Reviewer (NASA and NSF; 12 journals)

NASA and NSF grants. Journals: *Phys. Rev. Lett.*, *Phys. Rev. X*, *Phys. Rev. D*, *Rev. Mod. Phys.*, *Phys. Lett. B*, *Nature*, *Commun. Phys.*, *iScience*, *MNRAS*, *ApJ*, *ApJL*; guest editor for *PNAS*.

Additionally, I have internally refereed 28 articles for the LIGO Scientific Collaboration. I have also reviewed software for multiple analysis pipelines used in production analyses.

Seminar & Workshop Organizer

Gravitational waves study hub, Columbia	2026
Black-hole Ringdowns at CCA, Flatiron Institute	2022
LIGO Open Data Workshop, Flatiron Institute	2022
Pacific Coast Gravity Meeting (PCGM), Caltech	2018
Caltech/JPL Association for Gravitational-Wave Research (monthly seminars)	2015–2018

Columbia University

Graduate Admissions Committee (2025); Astrostatistics Major Design Committee (2025–present); Thesis Committees (Chengcheng Xin, Justin Vega, John Staunton); Columbia AI Verticals Project.

ADVISING &
MENTORING**PhD Students (2)**

Columbia: Abigail Moran, Ana Lam.

Undergraduate Students (4)

Columbia: Tata Tirapongprasert, Sarah Yuh, Candice Wu, Nathaniel Rodgers.

Postdoctoral Researchers (3)

Flatiron Institute: Asad Hussain, Konstantin Leyde, Keefe Mitman.

Students co-mentored before joining Columbia:

Graduate Students (16)

MIT: Ken Ng, Sylvia Biscoveanu, Yiwen Huang; *Caltech*: Simona Miller, Ethan Payne, Jacob Golomb, Rhiannon Udall, Keefe Mitman; *Columbia*: Harrison Siegel; *Stony Brook*: Soumendra Roy, Nicole Khusid; *Princeton*: Hengrui Zhu; *Minnesota*: Haowen Zhong; *Monash*: Teagan Clarke; *UT Austin*: Asad Hussain. *Montreal*: Ronan Legin.

Undergraduate Students (7)

Caltech: Carl Fields, Eve Chase, Praful Vasireddy, Jake Mattinson; *MIT*: Nishat Protyasha, Claire Williams; *Chinese U of Hong Kong*: Thomas Ng.

TEACHING

Instructor , Columbia University	2025–present
---	--------------

Modeling the Universe (ASTR 4260)

Cosmology, Black Holes, and Gravity (ASTR6014)

Guest Lecturer, MIT

2019

Physics course: Relativity (8.033); Introduction to Special Relativity (8.20); Selected Topics in Graduate Physics (8.398).

Summer Schools

<i>School on Astroparticle and Multi-Messenger Astrophysics</i> , São Paulo, Brazil	2026
Presenting a course on gravitational-wave astronomy at ICTP-SAIFR.	

International Center for Theoretical Physics, São Paulo, Brazil

2022

Summer school for South American teachers at ICTP-SAIFR (invited lecture).

International Institute of Physics, Natal, Brazil

2019

Structured and delivered one-week course on detecting and analyzing gravitational waves.

International School on Astroparticle Physics, Pierre Auger Observatory 2019
Invited lecturer on gravitational-wave detection and characterization.

Caltech GW Astronomy School, Caltech 2015
Assisted for course on gravitational waves from compact binaries (CGWAS 2015).

HONORS AND AWARDS

Fellowships & Scholarships

Flatiron Research Fellowship, Flatiron Institute, Simons Foundation 2021
Einstein Postdoctoral Fellowship, NASA Hubble Fellowship Program 2018
Summer Undergraduate Research Fellowship, California Institute of Technology 2012 & 2013
Board of Trustees Merit Scholarship, Loyola Marymount University 2010-2014
Kalbfleisch Physics Scholarship, Loyola Marymount University 2013-2014
John and Maria Laffin Scholarship, Loyola Marymount University 2011-2014

Presentation Awards

Everhart Lecture Award, California Institute of Technology 2018
Best talk at *Pacific Coast Gravity Meeting*, University of California, Santa Barbara 2017
Best poster at *GWPAW Conference*, Massachusetts Institute of Technology 2016
Best talk at *Karl Schwarzschild Meeting*, Frankfurt Institute of Advanced Studies, Germany 2013

Collaboration Awards

I have received the following prizes as a member of the LIGO Scientific Collaboration:

Princess of Asturias Award for Technical and Scientific Research 2017
UK Royal Astronomical Society 2017 Group Achievement Award in Astronomy 2017
Special Breakthrough Prize in Fundamental Physics 2016
Gruber Cosmology Prize, Yale University 2016

CERTIFICATIONS

NVIDIA Deep Learning Institute

1. [Fundamentals of Deep Learning](#)
2. [Data Parallelism: How to Train Deep Learning Models on Multiple GPUs](#)

Spoken Languages

Fluent: Spanish, English; *Intermediate*: French, Portuguese; *Basic*: German, Greek.

SOFTWARE



RINGDOWN: a Python package for analyzing black-hole ringdowns in gravitational wave data using Hamiltonian Monte-Carlo (<https://pypi.org/project/ringdown>).

JIM: accelerated and robust parameter estimation for gravitational waves using a JAX-based sampler (<https://pypi.org/project/jimGW>).

GWAXION: a Python package for computations related to gravitational wave emission by ultralight bosons, a dark matter candidate (<https://pypi.org/project/gwaxion>).

Others: I have contributed to LIGO-Virgo data analysis code in both Python and C++, including LALSuite (<https://git.ligo.org/lscsoft/lalsuite>) and Gwpy (<https://github.com/gwpy>).

The following are short-authorlist publications; see below for relevant LIGO and LISA Collaboration papers.

REFEREED PUBLICATIONS

mentees underlined
bold numeral denotes
≥ 50 citations
a star * denotes
≥ 100 citations

55. K. Mitman, **M. Isi**, and W. M. Farr, *Constraining Gravitational Wave Memory with Hierarchical Inference*, Phys. Rev. D, 114 (2026), p. 064016. [[arXiv:2605.27500](#)]
54. S. Bini, K. Krol, K. Chatziioannou, and **M. Isi**, *Impact of waveform systematics and Gaussian noise on the interpretation of GW231123*, Phys. Rev. D, 113 (2026), p. 083036. [[arXiv:2601.09678](#)]
53. H. Siegel, N. M. Khusid, **M. Isi**, and W. M. Farr, *GW231123 ringdown: interpretation as multimodal Kerr signal*, Phys. Rev. D, 114 (2026), p. 024050. [[arXiv:2511.02691](#)]



52. O. Gottlieb, B. D. Metzger, D. Issa, S. E. Li, M. Renzo, and **M. Isi**, *Spinning into the Gap: Direct-horizon Collapse as the Origin of GW231123 from End-to-end General-relativistic Magnetohydrodynamic Simulations*, *Astrophys. J. Lett.*, 993 (2025), p. L54. [[arXiv:2508.15887](#)]
51. C. Xin, **M. Isi**, W. M. Farr, and Z. Haiman, *Identifying Compact Chirping SMBHBs in LSST using Bayesian Analysis*, *Astrophys. J.*, 999 (2026), p. 149. [[arXiv:2506.10846](#)]
- ★ 50. E. Berti et al. inc. **M. Isi**, *Black hole spectroscopy: from theory to experiment*, *Class. Quantum Grav.*, 43 (2026), p. 123001. [[arXiv:2505.23895](#)]
49. S. J. Miller, **M. Isi**, K. Chatziioannou, V. Varma, and S. Hourihane, *Measuring spin precession from massive black hole binaries with gravitational waves: insights from time-domain signal morphology*, *Phys. Rev. D*, 112 (2025), p. 104046. [[arXiv:2505.14573](#)]
48. A. Hussain, **M. Isi**, and A. Zimmerman, *Hints of spin-magnitude correlations and a rapidly spinning subpopulation of binary black holes*, *Astrophys. J.*, 996 (2026), p. 71. [[arXiv:2411.02252](#)]
47. R. Legin, **M. Isi**, K. W. K. Wong, Y. Hezaveh, and L. Perreault-Levasseur, *Gravitational-wave Parameter Estimation in Non-Gaussian Noise Using Score-based Likelihood Characterization*, *Astrophys. J. Lett.*, 985 (2025), p. L46. [[arXiv:2410.19956](#)]
46. H. Siegel, **M. Isi**, and W. M. Farr, *Analyzing black-hole ringdowns II: data conditioning*, *Phys. Rev. D*, 111 (2025), p. 044070. [[arXiv:2410.02704](#)]
45. R. Udall, S. Hourihane, S. Miller, D. Davis, K. Chatziioannou, **M. Isi**, and H. Deshong, *Antialigned spin of GW191109: Glitch mitigation and its implications*, *Phys. Rev. D*, 111 (2025), p. 024046. [[arXiv:2409.03912](#)]
44. E. Payne, **M. Isi**, K. Chatziioannou, L. Lehner, Y. Chen, and W. M. Farr, *Curvature Dependence of Gravitational-Wave Tests of General Relativity*, *Phys. Rev. Lett.*, 133 (2024), p. 251401. [[arXiv:2407.07043](#)]
43. H. Zhong, **M. Isi**, K. Chatziioannou, and W. M. Farr, *Multidimensional hierarchical tests of general relativity with gravitational waves*, *Phys. Rev. D*, 110 (2024), p. 044053. [[arXiv:2405.19556](#)]
42. M. Lagos, L. Jenks, **M. Isi**, K. Hotokezaka, B. D. Metzger, E. Burns, W. M. Farr, S. Perkins, K. W. K. Wong, and N. Yunes, *Birefringence tests of gravity with multimessenger binaries*, *Phys. Rev. D*, 109 (2024), p. 124003. [[arXiv:2402.05316](#)]
41. T. Clarke, **M. Isi**, and others, *Toward a self-consistent framework for measuring black hole ringdowns*, *Phys. Rev. D*, 109 (2024), p. 124030. [[arXiv:2402.02819](#)]
40. H. Zhu, H. Siegel, K. Mitman, **M. Isi**, and others, *Black Hole Spectroscopy for Precessing Binary Black Hole Coalescences*, *Phys. Rev. D*, 111 (2025), p. 064052. [[arXiv:2312.08588](#)]
39. J. Golomb, **M. Isi**, and W. Farr, *Physical Models for the Astrophysical Population of Black Holes: Application to the Bump in the Mass Distribution of Gravitational Wave Sources*, *Astrophys. J.*, 976 (2024), p. 121. [[arXiv:2312.03973](#)]
38. R. Magee, **M. Isi**, E. Payne, K. Chatziioannou, W. M. Farr, G. Pratten, and S. Vitale, *Impact of selection biases on tests of general relativity with gravitational-wave inspirals*, *Phys. Rev. D*, 109 (2024), p. 023014. [[arXiv:2311.03656](#)]
37. **M. Isi** and W. M. Farr, *Comment on “Analysis of Ringdown Overtones in GW150914”*, *Phys. Rev. Lett.*, 131 (2023), p. 169001. [[arXiv:2310.13869](#)]
36. S. J. Miller, **M. Isi**, K. Chatziioannou, V. Varma, and I. Mandel, *GW190521: tracing imprints of spin-precession on the most massive black hole binary*, *Phys. Rev. D*, 109 (2024), p. 024024. [[arXiv:2310.01544](#)]
35. E. Payne, **M. Isi**, K. Chatziioannou, and W. M. Farr, *Fortifying gravitational-wave tests of general relativity against astrophysical assumptions*, *Phys. Rev. D*, 108 (2023), p. 124060. [[arXiv:2309.04528](#)] [**Editor’s suggestion**]
34. H. Siegel, **M. Isi**, and W. M. Farr, *Ringdown of GW190521: Hints of multiple quasinormal modes with a precessional interpretation*, *Phys. Rev. D*, 108 (2023), p. 064008. [[arXiv:2307.11975](#)]
33. T.C.K. Ng, **M. Isi**, K. W. Wong, and W. M. Farr, *Constraining gravitational wave amplitude birefringence with GWTC-3*, *Phys. Rev. D*, 108 (2023), p. 084068. [[arXiv:2305.05844](#)]

32. **M. Isi**, W. M. Farr, and V. Varma, *The directional isotropy of LIGO-Virgo binaries*, *Astrophys. J.*, 962 (2024), p. 19. [[arXiv:2304.13254](#)]
- ★ 31. K. W. Wong, **M. Isi**, and T. D. P. Edwards, *Fast gravitational wave parameter estimation without compromises*, *Astrophys. J.*, 958 (2023), p. 129. [[arXiv:2302.05333](#)]
30. T. D. P. Edwards, K. W. K. Wong, K. K. H. Lam, A. Coogan, D. Foreman-Mackey, **M. Isi**, and A. Zimmerman, *Differentiable and hardware-accelerated waveforms for gravitational wave data analysis*, *Phys. Rev. D*, 110 (2024), p. 064028. [[arXiv:2302.05329](#)]
29. **M. Isi**, *Parametrizing gravitational-wave polarizations*, *Class. Quantum Grav.*, 40 (2023), p. 203001. [[arXiv:2208.03372](#)]
28. M. Okounkova, **M. Isi**, K. Chatziioannou, and W. M. Farr, *Gravitational wave inference on a numerical-relativity simulation of a black hole merger beyond general relativity*, *Phys. Rev. D*, 107 (2023), p. 024046. [[arXiv:2208.02805](#)]
27. **M. Isi**, W. M. Farr, and K. Chatziioannou, *Comparing Bayes factors and hierarchical inference for testing general relativity with gravitational waves*, *Phys. Rev. D*, 106 (2022), p. 024048. [[arXiv:2204.10742](#)]
- ★ 26. V. Varma, S. Biscoveanu, T. Islam, F. H. Shaik, C.-J. Haster, **M. Isi**, W. M. Farr, S. E. Field, and S. Vitale, *Evidence of Large Recoil Velocity from a Black Hole Merger Signal*, *Phys. Rev. Lett.*, 128 (2022), p. 191102. [[arXiv:2201.01302](#)] [[Physics Magazine Synopsis](#)]
25. A. Vajpeyi, R. Smith, E. Thrane, G. Ashton, T. Alford, S. Garza, **M. Isi**, J. Kanner, T. J. Massinger, and L. Xiao, *A follow-up on intermediate-mass black hole candidates in the second LIGO–Virgo observing run with the Bayes Coherence Ratio*, *Mon. Not. Roy. Astron. Soc.*, 516 (2022), pp. 5309–5317. [[arXiv:2107.12109](#)]
24. V. Varma, **M. Isi**, S. Biscoveanu, W. M. Farr, and S. Vitale, *Measuring binary black hole orbital-plane spin orientations*, *Phys. Rev. D*, 105 (2022), p. 024045. [[arXiv:2107.09692](#)]
23. V. Varma, S. Biscoveanu, **M. Isi**, W. M. Farr, and S. Vitale, *Hints of spin-orbit resonances in the binary black hole population*, *Phys. Rev. Lett.*, 128 (2022), p. 031101. [[arXiv:2107.09693](#)]
22. S. Biscoveanu, **M. Isi**, V. Varma, and S. Vitale, *Measuring the spins of heavy binary black holes*, *Phys. Rev. D*, 104 (2021), p. 103018. [[arXiv:2106.06492](#)]
21. K. Chatziioannou, **M. Isi**, C.-J. Haster, and T. B. Littenberg, *Morphology-independent test of the mixed polarization content of transient gravitational wave signals*, *Phys. Rev. D*, 104 (2021), p. 044005. [[arXiv:2105.01521](#)]
20. **M. Isi**, *Recent LIGO-Virgo discoveries*, *Mod. Phys. Lett. A*, 36 (2021), p. 2130010. [[arXiv:2106.09529](#)] [[invited review](#)]
19. M. Okounkova, W. M. Farr, **M. Isi**, and L. C. Stein, *Constraining gravitational wave amplitude birefringence and Chern-Simons gravity with GWTC-2*, *Phys. Rev. D*, 106 (2022), p. 044067. [[arXiv:2101.11153](#)]
- ★ 18. **M. Isi**, W. M. Farr, M. Giesler, M. A. Scheel, and S. A. Teukolsky, *Testing the Black-Hole Area Law with GW150914*, *Phys. Rev. Lett.*, 127 (2021), p. 011103. [[arXiv:2012.04486](#)]
17. **M. Isi**, S. Mastrogiovanni, M. Pitkin, and O. J. Piccinni, *Establishing the significance of continuous gravitational-wave detections from known pulsars*, *Phys. Rev. D*, 102 (2020), p. 123027. [[arXiv:2010.12612](#)]
16. H.-Y. Chen, C.-J. Haster, S. Vitale, W. M. Farr, and **M. Isi**, *A Standard Siren Cosmological Measurement from the Potential GW190521 Electromagnetic Counterpart ZTF19abanrhr*, *Mon. Not. Roy. Astron. Soc.*, 513 (2022), pp. 2152–2157. [[arXiv:2009.14057](#)]
15. K.N.Y. Ng, **M. Isi**, C.-J. Haster, and S. Vitale, *Multiband gravitational-wave searches for ultralight bosons*, *Phys. Rev. D*, 102 (2020), p. 083020. [[arXiv:2007.12793](#)]
14. S. Biscoveanu, **M. Isi**, S. Vitale, and V. Varma, *New Spin on LIGO-Virgo Binary Black Holes*, *Phys. Rev. Lett.*, 126 (2021), p. 171103. [[arXiv:2007.09156](#)]
13. V. Varma, **M. Isi**, and S. Biscoveanu, *Extracting the Gravitational Recoil from Black Hole Merger Signals*, *Phys. Rev. Lett.*, 124 (2020), p. 101104. [[arXiv:2002.00296](#)]

- ★ 12. G. Vajente, Y. Huang, **M. Isi**, J. C. Driggers, J. S. Kissel, M. J. Szczepanczyk, and S. Vitale, *Machine-learning nonstationary noise out of gravitational-wave detectors*, Phys. Rev. D, 101 (2020), p. 042003. [[arXiv:1911.09083](#)]
 - ★ 11. L. Sun, R. Brito, and **M. Isi**, *Search for ultralight bosons in Cygnus X-1 with Advanced LIGO*, Phys. Rev. D, 101 (2020), p. 063020. [[arXiv:1909.11267](#)] [Erratum: [Phys. Rev. D 102, 089902 \(2020\)](#)]
 - ★ 10. **M. Isi**, M. Giesler, W. M. Farr, M. A. Scheel, and S. A. Teukolsky, *Testing the no-hair theorem with GW150914*, Phys. Rev. Lett., 123 (2019), p. 111102. [[arXiv:1905.00869](#)] [[Editor's suggestion](#)]
 - ★ 9. **M. Isi**, K. Chatziioannou, and W. M. Farr, *Hierarchical test of general relativity with gravitational waves*, Phys. Rev. Lett., 123 (2019), p. 121101. [[arXiv:1904.08011](#)]
 - ★ 8. M. Giesler, **M. Isi**, M. A. Scheel, and S. Teukolsky, *Black Hole Ringdown: The Importance of Overtones*, Phys. Rev. X, 9 (2019), p. 041060. [[arXiv:1903.08284](#)]
 - ★ 7. **M. Isi**, L. Sun, R. Brito, and A. Melatos, *Directed searches for gravitational waves from ultralight bosons*, Phys. Rev. D, 99 (2019), p. 084042. [[arXiv:1810.03812](#)] [Erratum: [Phys. Rev. D 102, 049901 \(2020\)](#)]
 - 6. **M. Isi** and L. C. Stein, *Measuring stochastic gravitational-wave energy beyond general relativity*, Phys. Rev. D, 98 (2018), p. 104025. [[arXiv:1807.02123](#)]
 - 5. **M. Isi**, R. Smith, S. Vitale, T. J. Massinger, J. Kanner, and A. Vajpeyi, *Enhancing confidence in the detection of gravitational waves from compact binaries using signal coherence*, Phys. Rev. D, 98 (2018), p. 042007. [[arXiv:1803.09783](#)]
 - ★ 4. T. Callister, [S. Biscoveanu](#), N. Christensen, **M. Isi**, A. Matas, O. Minazzoli, T. Regimbau, M. Sakellariadou, J. Tasson, and E. Thrane, *Polarization-based Tests of Gravity with the Stochastic Gravitational-Wave Background*, Phys. Rev. X, 7 (2017), p. 041058. [[arXiv:1704.08373](#)]
 - ★ 3. **M. Isi**, M. Pitkin, and A. J. Weinstein, *Probing Dynamical Gravity with the Polarization of Continuous Gravitational Waves*, Phys. Rev. D, 96 (2017), p. 042001. [[arXiv:1703.07530](#)]
 - 2. **M. Isi**, A. J. Weinstein, C. Mead, and M. Pitkin, *Detecting Beyond-Einstein Polarizations of Continuous Gravitational Waves*, Phys. Rev. D, 91 (2015), p. 082002. [[arXiv:1502.00333](#)]
 - 1. **M. Isi**, J. Mureika, and P. Nicolini, *Self-Completeness and the Generalized Uncertainty Principle*, JHEP, 11 (2013), p. 139. [[arXiv:1310.8153](#)]
- PREPRINTS
- 7. [A. Hussain](#), **M. Isi**, and A. Zimmerman, *Evidence for mass-dependent spin subpopulations in GWTC-4*, (2026). [[arXiv:2605.24281](#)]
 - 6. [N. M. Khusid](#), W. M. Farr, and **M. Isi**, *Polarization Analysis of Ringdown Signals*, (2026). [[arXiv:2605.15271](#)]
 - 5. [A. Hussain](#), **M. Isi**, and A. Zimmerman, *Living on the edge: Testing for compact population features at the edges of parameter space*, (2025). [[arXiv:2510.20010](#)] (submitted to ApJ)
 - ★ 4. **M. Isi** and W. M. Farr, *Revisiting the ringdown of GW150914*, (2022). [[arXiv:2202.02941](#)]
 - ★ 3. **M. Isi** and W. M. Farr, *Analyzing black-hole ringdowns*, (2021). [[arXiv:2107.05609](#)]
 - 2. **M. Isi** and A. J. Weinstein, *Probing gravitational wave polarizations with signals from compact binary coalescences*, (2017). [[arXiv:1710.03794](#)]
 - 1. M. Pitkin, **M. Isi**, J. Veitch, and G. Woan, *A nested sampling code for targeted searches for continuous gravitational waves from pulsars*, (2017). [[arXiv:1705.08978](#)]

PROCEEDINGS &
WORKSHOP
CONTRIBUTIONS

5. R. Legin, **M. Isi**, K. Wong, A. Adam, L. Perreault-Levasseur, and Y. Hezaveh, *Towards unbiased gravitational-wave parameter estimation using score-based likelihood characterization*, in 2nd ICML Workshop on Machine Learning for Astrophysics (ICML 2023), 2023. [<https://ml4astro.github.io/icml2023/assets/70.pdf>]
4. **M. Isi**, *Testing the no-hair theorem with LIGO and Virgo*, in Proceedings of the 3rd World Summit on Exploring the Dark Side of the Universe (EDSU2020): March 9–13, 2020, Guadeloupe Islands, 2020, pp. 49–58. [<https://kuscholarworks.ku.edu/handle/1808/30804>]
3. S. Köppel, M. Knipfer, **M. Isi**, J. Mureika, and P. Nicolini, *Generalized uncertainty principle and extra dimensions*, Springer Proc. Phys., 208 (2018), pp. 141–147. [[arXiv:1703.05222](https://arxiv.org/abs/1703.05222)]
2. R. Tso, **M. Isi**, Y. Chen, and L. Stein, *Modeling the Dispersion and Polarization Content of Gravitational Waves for Tests of General Relativity*, in Proceedings, 7th Meeting on CPT and Lorentz Symmetry (CPT 16): Bloomington, Indiana, USA, June 20–24, 2016, 2017, pp. 205–208. [[arXiv:1608.01284](https://arxiv.org/abs/1608.01284)]
1. **M. Isi**, J. Mureika, and P. Nicolini, *Self-completeness in Alternative Theories of Gravity*, Springer Proc. Phys., 170 (2016), pp. 95–101. [[arXiv:1402.3342](https://arxiv.org/abs/1402.3342)]

COLLABORATION
PUBLICATIONS

I am an author in 191 LIGO and Virgo Collaboration articles, listed [online](#). Out of those, I have led or made key contributions to the following:

12. A. G. Abac and others, *GWTC-5.0: Population Properties of Merging Compact Binaries*, (2026). [[arXiv:2605.27226](https://arxiv.org/abs/2605.27226)] I co-wrote the manuscript and led the analysis that revealed a subpopulation of high-spin black holes.
11. A. G. Abac and others, *GW250114: Testing Hawking’s Area Law and the Kerr Nature of Black Holes*, Phys. Rev. Lett., 135 (2025), p. 111403. [[arXiv:2509.08054](https://arxiv.org/abs/2509.08054)] I chaired the writing team and led the flagship analyses, based on techniques I established.
10. A. G. Abac and others, *GW231123: a Binary Black Hole Merger with Total Mass 190–265 $M_{\odot}$* , Astrophys. J. Lett., 993 (2025), p. L25. [[arXiv:2507.08219](https://arxiv.org/abs/2507.08219)] I contributed to the ringdown and systematics analyses.
9. R. Abbott and others, *Tests of General Relativity with GWTC-3*, Phys. Rev. D, 112 (2025), p. 084080. [[arXiv:2112.06861](https://arxiv.org/abs/2112.06861)] I contributed the infrastructure for combining observations from multiple events.
8. R. Abbott and others, *Tests of general relativity with binary black holes from the second LIGO-Virgo gravitational-wave transient catalog*, Phys. Rev. D, 103 (2021), p. 122002. [[arXiv:2010.14529](https://arxiv.org/abs/2010.14529)] [Erratum: [Phys. Rev. D 112, 049902 \(2025\)](https://arxiv.org/abs/2112.04990)]
I chaired this paper: orchestrated the results and led the writing.
7. R. Abbott and others, *GW190521: A Binary Black Hole Merger with a Total Mass of 150 $M_{\odot}$* , Phys. Rev. Lett., 125 (2020), p. 101102. [[arXiv:2009.01075](https://arxiv.org/abs/2009.01075)]
I carried out the preferred parameter estimation for this source, and contributed waveform systematic studies.
6. R. Abbott and others, *Properties and Astrophysical Implications of the 150 $M_{\odot}$ Binary Black Hole Merger GW190521*, Astrophys. J. Lett., 900 (2020), p. L13. [[arXiv:2009.01190](https://arxiv.org/abs/2009.01190)]
I produced the main parameter estimation and analyzed ringdown and recoil.
5. B. P. Abbott and others, *Tests of General Relativity with the Binary Black Hole Signals from the LIGO-Virgo Catalog GWTC-1*, Phys. Rev. D, 100 (2019), p. 104036. [[arXiv:1903.04467](https://arxiv.org/abs/1903.04467)]
I wrote several sections as part of the editorial team; contributed the polarizations analysis.
4. B. P. Abbott and others, *Tests of General Relativity with GW170817*, Phys. Rev. Lett., 123 (2019), p. 011102. [[arXiv:1811.00364](https://arxiv.org/abs/1811.00364)] I carried out and wrote the polarization analysis.
3. B. P. Abbott and others, *First narrow-band search for continuous gravitational waves from known pulsars in advanced detector data*, Phys. Rev. D, 96 (2017), p. 122006. [[arXiv:1710.02327](https://arxiv.org/abs/1710.02327)] [Erratum: [Phys. Rev. D 97, 129903 \(2018\)](https://arxiv.org/abs/1710.12990)]
I contributed a coherence study of detection outliers.
2. B. P. Abbott and others, *First search for nontensorial gravitational waves from known pulsars*, Phys. Rev. Lett., 120 (2018), p. 031104. [[arXiv:1709.09203](https://arxiv.org/abs/1709.09203)]
I was the lead author and analyst: performed the study and wrote the paper on behalf of the Collaborations.

1. B. P. Abbott and others, *GW170814: A Three-Detector Observation of Gravitational Waves from a Binary Black Hole Coalescence*, Phys. Rev. Lett., 119 (2017), p. 141101. [[arXiv:1709.09660](#)]
I contributed the polarization analysis and wrote the respective section.

I have contributed to the following LISA Consortium publications:

3. P. A. Seoane and others, *The Effect of Mission Duration on LISA Science Objectives*, Gen. Rel. Grav., 54 (2022), p. 3. [[arXiv:2107.09665](#)]
2. E. Barausse and others, *Prospects for Fundamental Physics with LISA*, Gen. Rel. Grav., 52 (2020), p. 81. [[arXiv:2001.09793](#)]
1. V. Baibhav and others, *Probing the nature of black holes: Deep in the mHz gravitational-wave sky*, Exper. Astron., 51 (2021), pp. 1385–1416. [[arXiv:1908.11390](#)]

INVITED TALKS
video links in blue
when available

55. *TBD*, International Symposium on Relativistic Astrophysics, Shanghai Apr 2027
54. *TBD*, RESCEU-NBIA Workshop, University of Tokyo Nov 2026
53. *TBD*, Stevens Institute of Technology [**colloquium**] Oct 2026
52. *Observing quasinormal modes and memory*, Galileo Galilei Center, Florence Sep 2026
51. *Ringdown observations*, YITP, Kyoto University June 2026
50. *The challenge and promise of LIGO's heavy black holes*, CITA Seminar, Toronto May 2026
49. *Learning about gravity with LIGO and Virgo*, University of Nevada, Las Vegas Apr 2026
48. *GWs: cosmology, ringdowns, memory*, Institute of Advanced Study Mar 2026
47. *Astrophysical black hole ringdowns*, RIKEN iTHEMS-Kyoto University Mar 2026
46. *The challenge and promise of LIGO's heavy black holes*, Astronomy Seminar, NYU Feb 2026
45. *GW250114: an exceptional detection*, GPPAW, Georgia Tech [**plenary**] Dec 2025
44. *Latest results from LIGO-Virgo-KAGRA*, COSMO-25, UPenn [**plenary**] Oct 2025
43. *Fundamental physics meets astrophysics*, University of Chicago Aug 2025
42. *A decade of GW astronomy*, Universidad de la República, Montevideo, Uruguay Jul 2025
41. *GWs from black holes to the cosmos*, Georgia Tech Nov 2024
40. *Cosmology with LIGO black holes*, National Observatory, Rio de Janeiro, Brazil Nov 2024
39. *GWs: from black holes to the cosmos*, Syracuse University [**colloquium**] Oct 2024
38. *The first decade of GW astronomy*, CosmoSul VII, Univ. Federal da Bahia, Brazil Aug 2024
37. *GW observations for fundamental fields*, [New Horizons for Psi Workshop](#), Lisbon July 2024
36. *Tests of GR with future ground-based detector*, [Dawn VII](#), UBC, Vancouver June 2024
35. *Physics with GWs from heavy black holes*, Gravity Seminar, Virginia University Mar 2024
34. *Gravitational-waves: from black holes to the cosmos*, Columbia University [**colloq.**] Feb 2024
33. *From black holes to the cosmos: physics with gravitational waves*, UC Berkeley Feb 2024
32. *Cosmology with GWs from black holes*, University of Rhode Island [**colloquium**] Feb 2024
31. *Extracting quasinormal modes from NR*, Gravity lunch talk, Cornell University Nov 2023
30. *The physics of black hole mergers*, Weinberg Seminar, University of Texas, Austin Oct 2023
29. *GW astronomy: from black holes to the cosmos*, CMU–Pittsburgh [**colloquium**] Oct 2023
28. *Probing black-hole spacetimes with GWs*, Physics Theory Seminar, Columbia May 2023
27. *GWs: digging the stellar graveyard*, Loyola Marymount University Nov 2023
26. *Black hole spectroscopy*, Princeton Gravity Initiative, Princeton University Oct 2022
25. *GW astronomy: black holes and beyond*, University of Illinois, Urbana-Champaign Oct 2022
24. *Stellar astrophysics with gravitational waves*, SACNAS Conference, Puerto Rico Oct 2022
23. *Astro and waveforms to test fund. physics*, GWs Beyond Boxes, Perimeter Institute Mar 2022

22. *Black hole ringdowns*, Black Hole Initiative, Harvard University **[colloquium]** Nov 2021
21. *Probing strong gravity with black-hole ringdowns*, LAW Physics, online Sep 2021
20. *Probing strong gravity with black-hole ringdowns*, LIGO Seminar, Caltech & MIT Aug 2021
19. *Learning about gravity with LIGO and Virgo*, Amaldi Conference, online **[plenary]** July 2021
18. *Testing GR with LIGO-Virgo*, Gravity seminar, SISSA & IFPU June 2021
17. *Probing strong gravity: BH ringdowns*, Strong-gravity seminar, Perimeter Institute May 2021
16. *Tests of general relativity with LIGO and Virgo*, APS April Meeting, online **[plenary]** Apr 2021
15. *Testing general relativity with GWTC-2*, University of British Columbia Feb 2021
14. *Testing GR with binary black holes from GWTC-2*, LIGO-Virgo webinar Nov 2020
13. *GWs and the nature of BHs*, Workshop: Gravity with EHT, Princeton [covid canceled] Aug 2020
12. *The discovery of GW190412*, APS April Meeting, Washington D.C. **[plenary]** Apr 2020
11. *Black-hole spectroscopy Summit on Exploring the Dark Universe*, Guadeloupe Mar 2020
10. *Testing the no-hair theorem with GW150914*, GWVerse COST action, Trieste Jan 2020
9. *Detecting GWs with LIGO*, GW Challenges and Cosmology IIP Workshop, Brazil June 2019
8. *Testing the no-hair theorem with GW150914*, VIA Lecture, IN2P3, CNRS, France May 2019
7. *High-frequency GW sources*, Gravitational Wave Advanced Detector Workshop, Italy May 2019
6. *Testing GR with gravitational waves*, KICP Seminar, University of Chicago Apr 2019
5. *Ondas Gravitacionales*, Instituto Tecnológico de Monterrey, Mexico Mar 2019
4. *Detecting new particles with gravitational waves*, BHI, Harvard **[colloquium]** Dec 2018
3. *Directed searches for GWs from ultralight bosons*, LIGO Seminar, Caltech Oct 2018
2. *Ondas gravitacionales: nueva ventana al universo*, UDELAR, Montevideo, Uruguay Sep 2016
1. *Testing GR with continuous GW polarizations*, LIGO Seminar, Caltech Oct 2014

OUTREACH

Media and Press Coverage

My work has been covered by a number of media outlets in the US and abroad. Highlights include:

28. **CNN**, *Black hole collision confirms predictions by Einstein and Hawking* Sep 11, 2025
27. **NPR**, *After 10 years of black hole science, Stephen Hawking is proven right* Sep 11, 2025
26. **Reuters**, *Astronomers get best view yet of two merging black holes* Sep 10, 2025
25. **New York Times**, *Happy Birthday, LIGO. Now Drop Dead.* Sep 10, 2025
24. **Astrobit.es**, *Uncovering Precession for GW190521* Jun 21, 2025
23. **Astrobit.es**, *The importance of astrophysics in tests of general relativity* Oct 17, 2023
22. **Popular Mechanics**, *What happens when two black holes collide* Nov 3, 2021
21. **New York Times**, *Did Death Cheat Stephen Hawking of a Nobel Prize?* Oct 4, 2021
20. **BBC News**, *La predicción de Stephen Hawking sobre los agujeros negros...* Jul 7, 2021
19. **Vice**, *Scientists Confirm Stephen Hawking's 50-Year-Old Theory...* July 2, 2021
18. **Daily Mail**, *Stephen Hawking is proved right* Jul 1, 2021
17. **Big Think**, *Black holes follow the laws of thermodynamics* June 23, 2021
16. **El País**, *Ecos del universo: joven uruguayo desentraña las ondas gravitacionales* Sep 2, 2020
15. **Nature News**, *This black-hole collision just made GWs even more interesting* Apr 20, 2020
14. **Science Magazine**, *Gravitational waves reveal unprecedented collision...* Apr 18, 2020
13. **InsideScience**, *LIGO Could See Black Holes Being Thrown Out of Galaxies* Apr 15, 2020
12. **Huffington Post**, *Les trous noirs seraient chauves, contrairement à ce que...* Sep 20, 2019
11. **Fox News**, *Stephen Hawking was wrong: Black Holes Are Bald* Sep 19, 2019

10. New Scientist (NL), [Zwart gat lijkt inderdaad geen haar te hebben](#) Sep 18, 2019
9. **New Scientist**, [Black hole that ‘rings’ like a bell shows Einstein was right](#) Sep 16, 2019
8. LiveScience [Hidden Gravitational Wave Signal Reveals that Black Holes Are Bald](#) Sep 17, 2019
7. **Boston Globe**, [Researchers detect gravitational waves from infant black hole](#) Sep 13, 2019
6. AstronomyNow, [Black hole’s gravitational ringing supports ‘no hair’ theorem](#) Sep 12, 2019
5. El Observador, [El uruguayo que desafía a Einstein](#) Oct 9, 2017
4. LIGO Documentary, [The new age of gravitational wave astronomy](#) July 19, 2017
3. ResearchGate Blog, [Expert dialogue: gravitational wave research](#) Mar 10, 2016
2. El País, [“Escuchan” fenómeno físico que predijo Albert Einstein](#) Feb 12, 2016
1. **Los Angeles Times**, [How scientists detected gravitational waves](#) Feb 11, 2016

I have appeared on Uruguayan radio and television, including:

4. Radio Carve, *A Ritmo de Tanco*: 2023 (radio)
3. Radio Mundo, *En Perspectiva*: 2016, 2017 and 2018 (radio)
2. Radio Del Sol, *No Toquen Nada*: 2017, 2018 and 2020 (radio)
1. Canal 4, *Telemundo* (evening news): 2016 (television)

Popular Writing

I have written a number of popular-level articles on gravitational waves including:

1. Caltech Letters, [A new window into the Universe](#) Sep 24, 2019
2. The Science Breaker, [Black holes ring like Einstein predicted](#) Oct 28, 2020
3. LIGO Lab website [LIGO and Virgo announce new detections in updated catalog](#) Oct 28, 2020
4. Astronomy Magazine, [Ask an astronomer: what would happen ...](#) Dec 1, 2020

Arts Engagement

I collaborate with artists and other creatives to explore the intersection of art and science.

1. *Dark Fringe*, with artist [Jules Gimbrone](#) sponsored by the Simons Foundation (see [coverage](#));
2. *Dancing with Black Holes*, with choreographer Peter Chu, composer Matthew Jamal, and sound designer Nick Ryan; conceived by Kay Matschullat (Brooklyn Academy of Music, April 2026);
3. *Astrometry through the ages*, with Prof. [Rana Adhikari](#) (Caltech) and digital creator [Carl Knox](#);
4. “Ringdown” at *Human Resources*, with artist [Alex Hawthorn](#) (see [performance](#)).

Community Outreach

I participate in science events at local schools, and give outreach talks to varied audiences (e.g., schoolchildren, teachers, political staffers), in both English and Spanish. Highlighted events:

21. *Ask a Scientist: Moon Tree Launch Party*, Madison Square Park, New York, NY Jul 16, 2026
20. *Ten Years of Gravitational Waves*, Italian Academy, Columbia University Dec 4, 2025
19. *A Signal in the Noise*, Simons Foundation, New York, NY Mar 14, 2025
18. *Let’s Talk Science: A Lens on space*, Simons Foundation, New York, NY Oct 5, 2023
17. *Creative Collisions: Science × Sound*, Simons Foundation, New York, NY Feb 16, 2023
16. *Elbio Fernández School*, Montevideo, Uruguay (virtual) Sep 8, 2022
15. *Mercedes High School*, Mercedes, Uruguay (virtual) Sep, 2 2021
14. *Paysandú Primary School*, Paysandú, Uruguay (virtual) Jun 11, 2020
13. *International Physics Symposium*, Monterrey, Mexico Mar 15, 2019
12. *Norma Coombs Elementary Science Literacy Night*, Pasadena, CA Mar 3, 2017
11. *Science Night Cleveland Elementary*, Pasadena, CA Jan 26, 2017

-
- | | |
|---|-----------------|
| 10. <i>Legislative Staffers Caltech Holiday Reception</i> , Pasadena, CA | Dec 1, 2016 |
| 9. <i>Washington STEAM Magnet School Outreach Night</i> , Pasadena, CA | Nov 9, 2016 |
| 8. <i>Colegio Seminario High-School</i> , Montevideo, Uruguay | Sept 30, 2016 |
| 7. <i>Iberoamerican High-school Physics Olympiads</i> , Carmelo, Uruguay | Sept 28, 2016 |
| 6. <i>High-school Physics Teachers Meeting</i> , Piriapolis, Uruguay | Sept 21, 2016 |
| 5. <i>Longfellow Elementary Stargazing and Science Night</i> , Pasadena, CA | Feb 26, 2016 |
| 4. <i>St. Philips Reverse Science Fair</i> , Pasadena, CA | Nov 6, 2015 |
| 3. <i>Caltech Community Science Event for Local Teachers</i> , Pasadena, CA | May 18, 2015 |
| 2. <i>Flintridge Prep Student Intern Day at Caltech</i> , Pasadena, CA | May 18, 2015 |
| 1. <i>SACNAS National Conference</i> , Los Angeles, CA | Oct 16–18, 2014 |